

Quantum & Crypto: Bet on the Migration, Not the Doomsday

An early-stage investment thesis · June 2026

The thesis

Quantum computing will not end crypto. It will force the largest cryptographic migration in its history — and that migration, not the doomsday, is the trade. Over the next three to five years the opportunity for an early-stage investor is not wagering on if or when a cryptographically relevant quantum computer (CRQC) arrives; that is a hardware coin-flip we cannot underwrite. It is the migration layer: the signatures, wallets, agility tooling and proofs that must carry trillions of dollars of value — and the base chains themselves — onto quantum-safe foundations while keeping trust intact.

This sits squarely in how we already think about crypto. Crypto exists to turn “trust me” into “verify it.” Quantum threatens the verifiable commitments that are crypto’s reason to exist; whoever rebuilds them on post-quantum foundations owns critical infrastructure for the decade after. The winners will not be the loudest about Q-Day — they will be the teams whose rails everyone is forced to migrate onto.

Why now — the gap between a closing threat and a long migration

The threat compressed sharply in 2026. In March, Google Quantum AI estimated that breaking the 256-bit elliptic-curve cryptography Bitcoin and Ethereum use to sign could require roughly 1,200 logical qubits — about 20× fewer than prior estimates. No CRQC exists today, and the largest machines (IBM’s ~120-qubit Nighthawk) are still generations away; credible Q-Day estimates span 2028 to the early 2030s. The date is not the point. The opportunity is the gap between an uncertain Q-Day and a migration that takes years — Bitcoin’s proposed signature sunset spans roughly five; Ethereum targets core post-quantum infrastructure by ~2029.

Two facts make this investable now rather than later. First, harvest-now-decrypt-later: exposed public keys are recordable today and breakable afterward — over a third of all Bitcoin already sits in quantum-exposed addresses, so the clock started before the computer exists. Second, the migration has visibly begun — Bitcoin merged BIP-360 (a quantum-resistant address type) in February 2026 and proposed BIP-361 to sunset legacy signatures in April; Ethereum stood up a dedicated post-quantum security team in January 2026, betting on account abstraction for signature agility and hash-based signatures for validators. The edge is to invest in the gap: after the threat stopped being science fiction, before it is fully priced.

Categories worth backing (3–5 years)

- 1. Crypto-agility middleware — highest conviction.** The transition is the product. Almost every chain, wallet and custodian hard-codes a single signature scheme, and swapping it without forks or downtime is largely unsolved. Tooling that rotates schemes, dual-signs through the transition and audits quantum exposure is the picks-and-shovels layer for a migration the whole industry is obligated to undertake — larger than the TLS/SSL migration, and with a hard deadline.
- 2. Quantum-safe wallets, signing & account abstraction.** The end-user surface and the likeliest land-grab. Account abstraction (Ethereum’s EIP-8141 path) turns signature agility into a product feature — users move to ML-DSA / FN-DSA or hash-based schemes without changing addresses. Quantum-safe-by-default wallets, signers and custody win the holders who are forced to act; Solana’s Winternitz vault and Ethereum’s leanXMSS validator path are early demand signals.
- 3. Harvest-now-decrypt-later mitigation & exposure intelligence — the near-term revenue.** Because the clock is already running, the first real budgets are defensive. Tools that quantify an entity’s quantum exposure (reused addresses, exposed public keys, mempool leakage) and proactively migrate

or shield value serve funds, custodians, exchanges and treasuries with capital at risk today — not in 2030.

4. Post-quantum × zero-knowledge — the deep-tech bet. ZK is core crypto infrastructure, and much of it must be made quantum-safe; hash-based and lattice proof systems are an open frontier, and the talent is exactly our hunting ground. These primitives also enable new mechanisms — such as the zero-knowledge proof-of-seed recovery Bitcoin is researching to rescue frozen coins. Longest horizon, highest defensibility.

5. Quantum-safe L1s / L2s — watch, mostly premature. New quantum-native chains are the obvious bet and, for this window, usually the wrong one: the incumbents are migrating in place, so a new L1 must win distribution it does not have on a threat that is not yet acute. Investable only where a chain earns a real application or community first and quantum-safety is a feature rather than the pitch.

What we'd avoid

We avoid the two crowded mistakes: quantum-doom narratives selling fear, and timing bets on hardware. Underwriting whether a CRQC lands in 2028 versus 2034 is speculation we cannot win, and any thesis that needs a specific Q-Day to pay off is fragile. We also avoid betting against the incumbents' in-place migration — Bitcoin and Ethereum are moving, so the money is in the rails that serve that migration, not in replacing them.

Conclusion

When the cryptography breaks, trust has to be rebuilt — and crypto is the trust layer. The teams that build the migration's rails — agility, quantum-safe signing, exposure intelligence, post-quantum-secure proofs — compound value through the transition and beyond. The doomsayers and the date-bettors do not. Bet on the migration.

Sources: NIST PQC standards (FIPS 203/204/205, Aug 2024; FN-DSA / FIPS 206 in development; HQC selected Mar 2025, draft 2026); Bitcoin BIP-360 (Feb 2026) and BIP-361 (Apr 2026); Ethereum Foundation post-quantum team (Jan 2026) and account-abstraction / leanXMSS roadmap; Google Quantum AI logical-qubit estimate (Mar 2026); IBM / Fujitsu hardware roadmaps. Figures are public estimates and forecasts — treat dates as ranges, not commitments.